# Locus-Level Clustering Reveals Distinct Subtypes of Genetic Pleiotropy Across Psychiatric Disorders, Including Antagonistic Effects Enriched for Cadherin Signaling

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**Abstract**

Cross-disorder genome-wide association studies have established that psychiatric disorders share substantial genetic architecture, but the locus-level structure of this sharing remains poorly characterized. Here, I extend the Grotzinger et al. (2026) cross-disorder Genomic SEM framework by applying unsupervised clustering to the five-dimensional z-score profiles of 611 genome-wide significant loci across five psychiatric genomic factors: compulsive, schizophrenia–bipolar, neurodevelopmental, internalizing, and substance use. Z-scores were sign-aligned to the internalizing factor to resolve arbitrary effect allele designation while preserving cross-factor directional relationships. K-means clustering identified three distinct subtypes of pleiotropy, corroborated by hierarchical clustering, Gaussian mixture models, and DBSCAN. Cluster 0 (355 loci) comprised broadly pleiotropic variants with concordant positive effects across all factors, dominated by internalizing signal. Cluster 1 (182 loci) captured psychotic-spectrum-specific loci driven by schizophrenia–bipolar effects. Cluster 2 (74 loci) revealed antagonistic pleiotropy, with positive internalizing effects coupled with negative schizophrenia–bipolar, compulsive, and substance use effects. This antagonistic cluster showed the strongest enrichment for QSNP-significant loci (81.1%, $\chi^2$ $p = 7 \times 10^{-54}$) and was the only cluster with significant pathway enrichment, converging on cadherin and cell-cell adhesion signaling (GO:BP, KEGG; $p = 1.26 \times 10^{-5}$). These results suggest that the shared genetic architecture of psychiatric disorders is not uniform but organizes into qualitatively distinct modes, including a biologically coherent set of loci that differentially affect internalizing and psychotic-spectrum conditions through cell adhesion mechanisms.

**Keywords:** psychiatric genomics, pleiotropy, GWAS, Genomic SEM, unsupervised clustering, antagonistic pleiotropy, cadherin signaling, cross-disorder genetics

# 1 Introduction

Psychiatric disorders are among the most heritable complex traits, yet their genetic architectures are characterized by extensive pleiotropy. Pleiotropy refers to the phenomenon in which individual genetic variants influence multiple conditions simultaneously [Lee et al., 2019, Anttila et al., 2018]. Cross-disorder genome-wide association studies (GWAS) have consistently demonstrated that disorders such as schizophrenia, bipolar disorder, major depressive disorder, and attention-deficit/hyperactivity disorder share substantial genetic overlap, challenging traditional diagnostic boundaries [Lee et al., 2019].

Grotzinger et al. (2026) advanced this understanding by applying Genomic structural equation modeling (Genomic SEM) to GWAS summary statistics spanning 14 psychiatric disorders, identifying five latent genomic factors: compulsive (F1), schizophrenia–bipolar (F2), neurodevelopmental (F3), internalizing (F4), and substance use (F5) [Grotzinger et al., 2026]. This factor structure provides a parsimonious representation of the shared genetic architecture, and the study characterized 611 genome-wide significant loci on a factor-by-factor basis. Additionally, the study computed heterogeneity statistics ($Q_P$) to identify loci whose effects deviate significantly from the factor model's predictions.

However, the original analysis examined loci through the lens of individual factors, asking which loci are significant for each factor. This leaves open a complementary question: when loci are characterized by their full cross-factor effect profiles, do structured subtypes of pleiotropy emerge? Rather than classifying a locus as "an F2 hit" or "an F4 hit," this analysis asks whether the joint pattern of effects across all five factors reveals distinct modes of genetic influence on psychiatric illness. Understanding whether pleiotropy organizes into structured subtypes has implications for interpreting shared genetic liability, because different modes of pleiotropy (concordant vs. antagonistic) may reflect distinct biological mechanisms and could have different clinical implications for comorbidity and treatment.

This project addresses that question by constructing a five-dimensional z-score vector for each of the 611 genome-wide significant loci and applying unsupervised clustering to identify structured patterns of shared and opposing genetic effects. The goal is to characterize the locus-level organization of psychiatric pleiotropy and to determine whether computationally derived subtypes correspond to meaningful biological distinctions.

# 2 Data

All summary statistics were obtained from the Psychiatric Genomics Consortium (PGC) under their standard data use terms. Raw data files are not redistributed in accordance with PGC policy.

## 2.1 Primary Data

The primary input consisted of factor-level GWAS summary statistics from the 14-disorder Genomic SEM model reported in Grotzinger et al. (2026) [Grotzinger et al., 2026]. Five factor-level summary statistic files were used, corresponding to the compulsive (F1), schizophrenia–

bipolar (F2), neurodevelopmental (F3), internalizing (F4), and substance use (F5) genomic factors. Each file provides per-SNP effect estimates ($\beta$, SE) for the respective factor.

### 2.2 Locus Selection

The analysis focused on 611 unique lead SNPs identified as genome-wide significant ($p < 5 \times 10^{-8}$) in the original study. These loci reached significance on at least one of the five factors, the hierarchical p-factor, or the associated heterogeneity (Q) tests. The full set of 611 loci was used regardless of which specific test produced significance, ensuring comprehensive coverage of the pleiotropic landscape.

### 2.3 Quality Control

All SNPs were filtered to imputation quality INFO $> 0.9$. The 611 lead SNPs were intersected across all five factor-level summary statistic files to ensure consistent allele representation. All 611 lead SNPs were retained after filtering, with no missing values across any factor. SNPs present in all five files after filtering constituted the final analysis set.

## 3 Methods

### 3.1 Effect Size Standardization

For each of the 611 lead SNPs, z-scores were computed as $z = \beta/\mathrm{SE}$ from each of the five factor-level summary statistic files, producing a $611 \times 5$ effect matrix. Z-scores serve as the standard metric for cross-GWAS comparisons because they normalize for differences in sample size and allele frequency across the disorder-specific GWAS that contribute to each factor [Marees et al., 2018].

### 3.2 P-Factor Exclusion

The original Genomic SEM model also provides estimates for a hierarchical p-factor representing general psychopathology liability. Preliminary analysis revealed a Pearson correlation of 0.974 between p-factor z-scores and F4 (internalizing) z-scores across the 611 loci, indicating near-perfect multicollinearity. Including the p-factor as a sixth clustering dimension would have effectively double-weighted internalizing signal. The p-factor was therefore excluded from the clustering features, and the analysis proceeded with five dimensions. The p-factor still contributed to locus selection (Section 2.2), as some loci reached genome-wide significance only on the hierarchical p-factor test.

### 3.3 Allele Direction and Sign-Alignment

GWAS summary statistics report effect sizes relative to an arbitrarily designated effect allele. For a given SNP, the sign of the z-score reflects whether the effect allele increases or decreases liability for a given factor, but the choice of which allele is designated as the effect allele is arbitrary. In a single-trait GWAS, this is inconsequential. In a cross-factor analysis, however, the arbitrary allele designation means that the raw z-score vector for a locus can be globally sign-flipped without changing its biological meaning. For example, a locus with the profile $(+, +, -, +, -)$ is biologically identical to one with $(-, -, +, -, +)$ if the only difference is which allele was labeled as the effect allele.

This presents a methodological challenge for clustering. Preliminary analysis using raw (unaligned) z-scores produced a dominant two-cluster solution (silhouette = 0.462) that separated loci based on the sign of their effect allele rather than the biological pattern of effects. This was an artifact of the arbitrary allele designation, not a biological finding.

Three approaches were evaluated to address this:

1. **Raw z-scores:** Retained allele direction as given. Produced artifactual mirror-image clusters.

2. **Absolute values:** Removed all directional information ($|z|$). Eliminated the artifact but also eliminated the ability to detect antagonistic pleiotropy, as loci with opposing effects across factors would be indistinguishable from concordant ones.

3. **Sign-alignment to F4 (Internalizing):** For each SNP, if the F4 z-score was negative, all five z-scores were multiplied by $-1$, ensuring F4 was always non-negative. This preserved the relative directional relationships across factors (including antagonistic patterns) while resolving the arbitrary allele designation.

Sign-alignment to F4 was selected as the primary approach because it preserves the cross-factor directional structure necessary to detect antagonistic pleiotropy while resolving the allele-direction artifact. F4 (Internalizing) was chosen as the reference factor because it showed the strongest and most consistent signal across the locus set.

## 3.4 Standardization

After sign-alignment, the $611 \times 5$ effect matrix was standardized to zero mean and unit variance per factor using scikit-learn's StandardScaler. Standardization ensures that factors with larger absolute z-scores (e.g., F4) do not dominate the Euclidean distance calculations used in clustering.

## 3.5 Clustering

### 3.5.1 Primary Analysis: K-Means Clustering

K-means clustering was performed with 50 random initializations on the standardized, sign-aligned effect matrix. K-means partitions loci into $k$ clusters by iteratively assigning each locus to its nearest centroid and recomputing centroid positions to minimize total within-cluster variance. The 50 initializations ensure robustness to starting conditions by repeating the optimization from different random starting positions and retaining the solution with the lowest total within-cluster variance. K-means was chosen as the primary method because it is well-suited to continuous effect profiles in moderate dimensions and its objective function (within-cluster variance minimization) directly aligns with the goal of identifying groups of loci with similar cross-factor effect patterns.

### 3.5.2 Number of Clusters

The number of clusters ($k$) was evaluated from $k = 2$ to $k = 10$ using two criteria:

- **Silhouette score:** Measures how similar each point is to its own cluster compared to neighboring clusters. Ranges from $-1$ to $+1$; higher values indicate better-defined clusters.
- **Bootstrap stability:** 1,000 bootstrap resamples of the locus set were drawn, clustered at each $k$, and the adjusted Rand index (ARI) between the bootstrap and full-sample cluster assignments was computed. Higher mean ARI indicates more robust cluster structure.

### 3.5.3 Comparison Methods

Three additional clustering methods were run at matched $k$ values to assess robustness of the primary k-means solution:

- **Hierarchical agglomerative clustering:** Performed using Ward's minimum variance linkage with Euclidean distance. Ward linkage minimizes within-cluster variance at each merge step, producing compact clusters. Unlike k-means, hierarchical clustering is deterministic and produces a nested hierarchy of solutions via a dendrogram.
- **Gaussian mixture models (GMM):** Fit with full covariance matrices and 10 initializations, allowing each cluster to have an independent shape and orientation. Evaluated by Bayesian Information Criterion (BIC) and silhouette score.
- **DBSCAN:** Density-based clustering evaluated across a range of $\epsilon$ values (0.5–2.5) with minimum samples of 5. Used as a sanity check for non-spherical or density-based structure in the data.

All three comparison methods recovered qualitatively similar cluster profiles at $k = 3$, including a broadly pleiotropic cluster, a psychotic-spectrum-specific cluster, and an antagonistic pleiotropy cluster with positive internalizing and negative schizophrenia–bipolar effects, supporting the robustness of the primary k-means solution.

## 3.6 Cluster Validation with $Q_P$ Heterogeneity Statistics

The Genomic SEM framework produces, for each SNP and each factor, a $Q_P$ statistic quantifying the degree to which that SNP's effect deviates from the model's predicted factor loading pattern. Higher $Q_P$ values indicate greater heterogeneity relative to the latent factor structure. Additionally, the original study flagged loci reaching genome-wide significance ($p < 5 \times 10^{-8}$) on the Q test as "Q hits," indicating substantial deviation from the factor model.

$Q_P$ values were used for post-hoc cluster annotation only and were not included as clustering features. Two validation analyses were conducted:

1. **$Q_P$ gradient:** Mean and median $Q_P$ values were compared across clusters for each factor using the Kruskal–Wallis test (non-parametric, appropriate given skewed $Q_P$ distributions).

2. **Q hit enrichment:** The proportion of QSNP-significant loci ("Q hits") in each cluster was tested for differential enrichment using a $\chi^2$ test.

### 3.7 Pathway Enrichment Analysis

SNPs in each cluster were mapped to genes using g:Profiler's SNP-to-gene conversion functionality. Per-cluster gene sets were tested for enrichment against Gene Ontology Biological Process (GO:BP) and KEGG pathway databases. The background gene set was defined as the full set of genes mapped from all 611 loci (484 unique genes), rather than the genome-wide background, to control for the ascertainment bias inherent in using only genome-wide significant hits.

## 4 Results

### 4.1 Cluster Selection

Silhouette scores across $k = 2$ to $k = 10$ were highest at $k = 2$ (0.254) and $k = 3$ (0.256), with scores declining for $k \geq 4$. GMM BIC values favored $k = 4$ (8110.1) over $k = 2$ (8320.4) and $k = 3$ (8150.2), suggesting potential additional substructure. DBSCAN did not identify well-separated density-based clusters at any $\epsilon$ value tested, consistent with the continuous, overlapping nature of genetic effect profiles.

Given that $k = 2$ and $k = 3$ had nearly identical silhouette scores but $k = 3$ revealed a biologically distinct cluster not visible at $k = 2$ (see Section 4.2), $k = 3$ was selected as the primary result. The $k = 4$ solution was examined as a secondary analysis.

### 4.2 Three Subtypes of Pleiotropy at $k = 3$

Table 1 presents the mean z-scores for each cluster across the five factors.

Table 1: Mean z-scores per cluster at $k = 3$ (sign-aligned to F4 Internalizing). All 611 genome-wide significant loci from Grotzinger et al. (2026) are included.

| Cluster | $n$ loci | F1 Comp. | F2 SCZ-BIP | F3 Neurodev. | F4 Internal. | F5 Subst. |
|---|---|---|---|---|---|---|
| C0 | 355 | 1.24 | 2.82 | 1.30 | 6.23 | 2.20 |
| C1 | 182 | 1.54 | 5.98 | −0.09 | 2.42 | 1.64 |
| C2 | 74 | −0.51 | −3.50 | 0.06 | 2.26 | −0.77 |

#### 4.2.1 Cluster 0: Broadly Pleiotropic, Internalizing-Dominant (355 loci)

The largest cluster showed positive mean z-scores across all five factors, with the strongest signal on F4 (Internalizing, mean $z = 6.23$). These loci exert concordant risk-increasing effects across psychiatric domains, consistent with a general liability to psychopathology concentrated in internalizing conditions. This cluster contained 200 of 202 hierarchical p-factor hits and 172 of 179 F4-significant hits, confirming its alignment with broadly acting genetic variants.

### 4.2.2 Cluster 1: Psychotic-Spectrum Specific (182 loci)

This cluster was dominated by F2 (schizophrenia–bipolar, mean $z = 5.98$), with near-zero neurodevelopmental signal (F3, mean $z = -0.09$) and moderate internalizing effects (F4, mean $z = 2.42$). The negligible F3 signal indicates specificity to adult-onset psychotic illness rather than neurodevelopmental conditions. This cluster contained 89 of 125 F2-significant hits.

### 4.2.3 Cluster 2: Antagonistic Pleiotropy (74 loci)

The most novel finding was a 74-locus cluster exhibiting antagonistic pleiotropy: positive internalizing effects (F4, mean $z = 2.26$) paired with negative schizophrenia–bipolar (F2, mean $z = -3.50$), compulsive (F1, mean $z = -0.51$), and substance use (F5, mean $z = -0.77$) effects. These loci increase genetic risk for internalizing disorders (depression, anxiety, PTSD) while decreasing risk for psychotic-spectrum and externalizing conditions.

**Figure 1:** Mean z-score profiles for each cluster at $k = 3$. Bar plots showing the mean signed z-score across the five psychiatric genomic factors for each cluster. Cluster 0 shows concordant positive effects; Cluster 1 shows F2 dominance; Cluster 2 shows the antagonistic pattern with opposing F4 and F2 effects.

Figure 1: Cross-factor z-score profiles for three pleiotropy subtypes.

## 4.3 $Q_P$ Heterogeneity Validation

Clusters showed a monotonic $Q_P$ gradient across all five factors (Kruskal–Wallis $p < 0.001$ for each factor): Cluster 0 had the highest mean $Q_P$ values ($\sim$0.45), Cluster 1 intermediate values ($\sim$0.33), and Cluster 2 the lowest ($\sim$0.27). Lower $Q_P$ indicates greater deviation from the factor model's predictions.

Enrichment of QSNP-significant loci across clusters was highly significant ($\chi^2$ $p = 7.03 \times 10^{-54}$). The proportion of Q hits increased monotonically: Cluster 0 = 7.3%, Cluster 1 = 59.3%, Cluster 2 = 81.1%. The antagonistic pleiotropy cluster (C2) contained the highest concentration of loci whose effects deviate most from the Genomic SEM factor predictions, consistent with the interpretation that these loci violate the assumed factor structure in a structured, biologically meaningful way.

**Figure 2:** $\mathrm{Q_P}$ heterogeneity statistics by cluster. (A) Mean $\mathrm{Q_P}$ values per factor per cluster showing the monotonic gradient. (B) Proportion of QSNP-significant hits per cluster.

Figure 2: $\mathrm{Q_P}$ heterogeneity validation of cluster structure.

### 4.4 Pathway Enrichment

Pathway enrichment analysis using g:Profiler revealed significant enrichment exclusively in Cluster 2 (antagonistic pleiotropy). Clusters 0 and 1 showed no significant enrichment against the within-study gene background of 484 genes.

Table 2: Significant pathway enrichment terms for Cluster 2 (antagonistic pleiotropy, 71 mapped genes). Background: 484 genes mapped from all 611 loci.

| **Term** | **Source** | **Adjusted** $p$ | **Genes** |
|---|---|---|---|
| Homophilic cell-cell adhesion | GO:BP | $1.26 \times 10^{-5}$ | 13 |
| Cadherin signaling pathway | KEGG | $1.35 \times 10^{-5}$ | 13 |
| Cell-cell adhesion | GO:BP | $1.37 \times 10^{-3}$ | 17 |
| Cell adhesion | GO:BP | $1.07 \times 10^{-2}$ | 18 |

The convergence on cadherin and cell adhesion pathways is notable because cadherins are a family of proteins critically involved in neural circuit formation, synapse assembly, and neurodevelopmental processes. The enrichment of these pathways specifically in the antagonistic pleiotropy cluster, combined with their absence in the broadly pleiotropic and psychotic-specific clusters, suggests that cadherin-mediated cell adhesion may differentially influence mood-spectrum and psychotic-spectrum liability in opposing directions.

**Figure 3:** Pathway enrichment results by cluster. Only Cluster 2 (antagonistic pleiotropy) shows significant enrichment, converging on cell adhesion and cadherin signaling pathways.

Figure 3: Per-cluster pathway enrichment analysis.

### 4.5 Secondary Analysis at $k = 4$

Increasing to $k = 4$ subdivided the broadly pleiotropic Cluster 0 into two subgroups: an internalizing–substance subgroup (high F4 + high F5) and an internalizing–neurodevelopmental subgroup (high F4 + high F3, with F3 mean $z = 2.04$ representing the highest neurodevelopmental signal observed in any cluster). The psychotic-spectrum cluster (C1) and antagonistic pleiotropy cluster (C2) remained stable between $k = 3$ and $k = 4$, supporting the robustness of these structures.

### 4.6 Methodological Summary

**Figure 4:** Analysis workflow. From locus selection through effect size standardization, sign-alignment, clustering, and validation.

Figure 4: Analysis workflow and key design decisions.

# 5 Discussion

## 5.1 Summary of Findings

This analysis demonstrates that the shared genetic architecture of psychiatric disorders, as captured by the Grotzinger et al. (2026) Genomic SEM framework, organizes into qualitatively distinct modes of pleiotropy at the locus level. Three subtypes were identified: broadly pleiotropic loci with concordant effects across all factors, psychotic-spectrum-specific loci driven primarily by schizophrenia–bipolar signal, and a novel set of loci exhibiting antagonistic pleiotropy between internalizing and psychotic-spectrum disorders.

The antagonistic pleiotropy cluster is the most notable finding. These 74 loci increase risk for depression, anxiety, and PTSD while simultaneously decreasing risk for schizophrenia, bipolar disorder, and related conditions. This pattern was validated by two independent lines of evidence: the $\mathrm{Q_P}$ heterogeneity gradient, which confirmed that these loci showed the greatest deviation from the factor model's predictions, and pathway enrichment in cadherin and cell-cell adhesion signaling, a biologically plausible mechanism given the role of cadherins in neural circuit formation.

## 5.2 Relationship to Prior Work

The original Grotzinger et al. (2026) analysis characterized loci on a factor-by-factor basis, identifying which factors each locus reached significance on and which loci showed heterogeneous effects (Q hits). The present analysis complements this by showing that the cross-factor effect profiles organize into structured subtypes rather than a continuum. The $\mathrm{Q_P}$ validation demonstrates that these subtypes align with the original study's heterogeneity measures: the broadly pleiotropic cluster corresponds to loci that fit the factor model well, while the antagonistic cluster corresponds to loci that deviate most strongly.

The enrichment of cadherin signaling in the antagonistic cluster adds a biological dimension not present in the original factor-level characterization. While cell adhesion pathways have been previously implicated in psychiatric GWAS broadly, the present analysis localizes this signal to a specific subset of loci with a distinctive pattern of opposing cross-disorder effects. Antagonistic pleiotropy has been extensively studied in evolutionary genetics [Williams, 1957], and individual loci with opposing effects across psychiatric disorders have been identified in prior cross-disorder analyses [Lee et al., 2019]. However, the present analysis is, to our knowledge, the first to show that such loci organize into a structured subtype with coherent pathway biology rather than representing isolated exceptions to a generally concordant genetic architecture.

## 5.3 Limitations

Several limitations should be noted. First, the silhouette scores for the primary $k = 3$ solution (0.256) indicate modest but real cluster separation. This is expected given the continuous and overlapping nature of genetic effect profiles, but it means the cluster boundaries are not sharp. Second, the sign-alignment procedure anchors all loci to positive F4, which may introduce bias if F4 effects are systematically different from other factors; however, the consistency of the results across validation analyses ($Q_P$, pathway enrichment) suggests the clusters reflect genuine biological structure rather than alignment artifacts. Third, pathway enrichment was conducted using g:Profiler's SNP-to-gene mapping, which assigns each SNP to the nearest gene; more sophisticated gene mapping approaches (e.g., eQTL-based or chromatin interaction-based) might refine the enrichment results. Fourth, the analysis is based on summary statistics from European-ancestry GWAS, limiting generalizability to other populations.

### 5.4 Planned Extensions

Several additional analyses are planned to strengthen and extend these findings:

- **LD score regression:** To quantify genome-wide genetic correlations between psychiatric disorders using individual disorder-level GWAS summary statistics, providing an independent characterization of the cross-disorder genetic landscape.
- **Individual disorder-level analysis:** Repeating the clustering using z-scores from the 14 individual disorder GWAS (rather than the five Genomic SEM factors) as a sensitivity analysis, with and without underpowered disorders (OCD, Tourette syndrome).
- **Bayesian shrinkage estimates:** Using shrinkage-based effect sizes as an alternative to raw z-scores to assess whether the cluster structure is robust to effect size estimation approach.

## 6 Conclusion

Unsupervised clustering of locus-level cross-factor effect profiles reveals that psychiatric genetic pleiotropy is not a single phenomenon but encompasses at least three qualitatively distinct modes. The identification of a biologically coherent antagonistic pleiotropy cluster, enriched for cadherin signaling and characterized by opposing effects on internalizing and psychotic-spectrum disorders, suggests that the genetic relationship between psychiatric conditions is more complex than shared liability models alone would predict. These findings provide a complementary perspective to the factor-level characterization reported in the original cross-disorder Genomic SEM analysis and may inform future investigations into the biological mechanisms underlying psychiatric comorbidity and the potential for disorder-specific therapeutic targets.

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